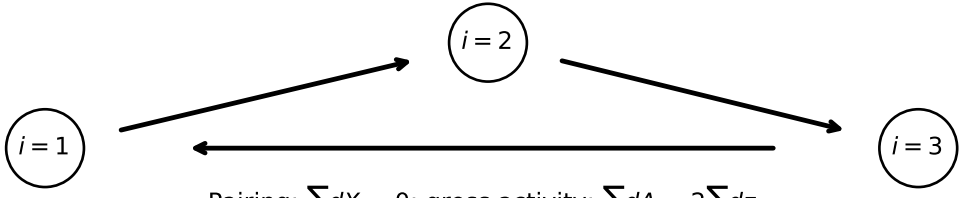

Pairing: $\sum_i dX_i = 0$; gross activity: $\sum_i dA_i = 2\sum_e dz_e$

$$dH = -\kappa \sum_i [S_i(S_i - F)/g(S_i)]\, dA_i \text{ (not conserved)}$$

C_0: simultaneous opposing flows allowed; C_1: no instantaneous opposing flow